

TECHNICAL AUDIT

Adult Census Income Algorithmic Decision-Support System

Auditing fairness, robustness, transparency, and privacy in a binary income classifier

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May 2026

Why audit this system?



THE TASK

Predict whether annual income > \$50K

This ADS uses socio-demographic and employment features from the Adult Census Income dataset to classify individuals into one of two income categories: > \$50K or ≤ \$50K

The dataset contains **32,561 records** from the 1994 Census Bureau database

Note on the \$50K threshold: The \$50K cutoff is not chosen by us. It is the predefined historical target label in the Adult Census Income dataset. Therefore, it should not be interpreted as a universal definition of being “rich”

Income category is often used as a proxy for **socio-economic status**. In settings such as **public policy, social science research, resource allocation, marketing, or other decision-support systems**, income predictions can shape how populations are analyzed and how services are targeted.



WHY IT MATTERS

Prediction errors can misrepresent a person's socio-economic status.

- 1 Whose income status does the model predict correctly?
- 2 Whose income status does the model misclassify?
- 3 Are errors concentrated in socially meaningful subgroups?

A model can have strong aggregate accuracy while still producing unequal errors across groups.

Dataset is predictive, but structurally imbalanced

INPUTS

14 features: The ADS uses socio-demographic and employment-related variables from the Adult Census Income dataset

Numerical (6)

age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week

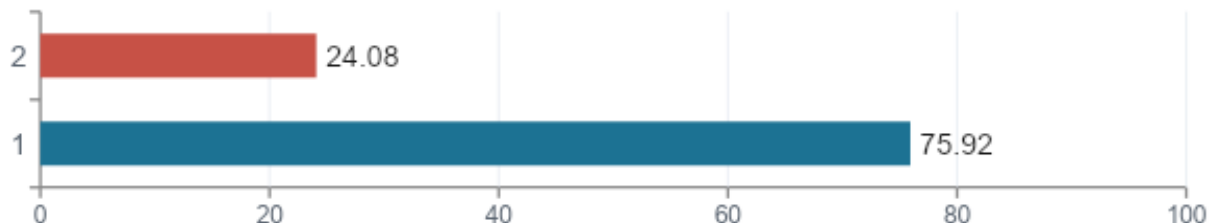
- Age and hours-per-week follow relatively normal distributions, with a large spike at 40 hours per week.
- Capital-gain and capital-loss are extremely right-skewed and zero-inflated; over 90% of individuals report zero values.
- Non-zero capital-gain and capital-loss values are strong predictors of high income.
- fnlwgt is a continuous demographic sampling weight and does not have a logical causal relationship with individual income

Categorical (8)

workclass, education, marital-status, occupation, relationship, race, sex, native-country

- Most categorical features show severe class imbalance, reflecting the socio-economic structure of 1994.
- Race is predominantly White, about 85%.
- Sex is skewed toward Male, about 67%.
- Native-country is overwhelmingly United States, about 90%.
- Standard missing-value checks report no nulls, but missing values are hidden as "?".
- Hidden missingness appears in workclass with 1,836 missing values, occupation with 1,843, and native-country with 583.

OUTPUT — Binary label, but the classes are imbalanced



The target class is highly imbalanced:

- **≤50K:** 24,720 people, **75.92%**
- **>50K:** 7,841 people, **24.08%**

This imbalance may lead to disparities in the model's **False Negative Rate** and **False Positive Rate**.

Original implementation: high accuracy, flawed validation



THE MODEL

Logistic Regression

- The original author trained a Logistic Regression model to predict whether an individual's income is $>50K$ or $\leq 50K$.
- The implementation also used randomized cross-validation search for hyperparameter tuning.
- The model outputs a binary predicted class label, although Logistic Regression can also be interpreted through predicted probabilities and learned coefficients.

Three preprocessing red flags

Drops race feature

"Fairness through unawareness" — proxies still leak race information.

Mode-imputes hidden "?"

Replaces missing workclass / occupation values with majority categories.

IQR age filter (>78)

IQR age filter excludes 166 older records and narrows the represented population.



VALIDATION RED FLAG

85% accuracy hides the real issue

Author's confusion matrix shows asymmetric errors:

664

False Negatives

$> \$50K$ predicted as $\leq \$50K$

321

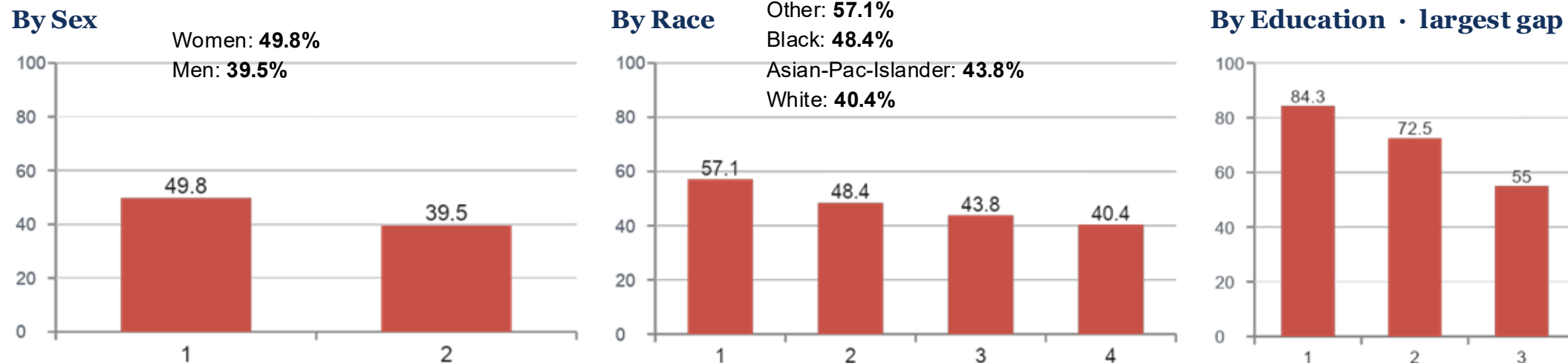
False Positives

$\leq \$50K$ predicted as $> \$50K$

$FN > 2 \times FP$ — the model biases toward the majority class to inflate accuracy.

Whose high incomes does the model miss?

False Negative Rate (FNR) — % of true >\$50K earners predicted as ≤\$50K



KEY INSIGHT

Aggregate accuracy is misleading. Women have **higher accuracy** (92.0% vs 81.7%) but **higher FNR** — the model misses more high-earning women. Disparities by race **remain even after race is removed** from training. The model leans heavily on formal education and misses high earners outside that profile.

The model does not satisfy group-fairness criteria

Statistical Parity

Gap in predicted-positive rate across groups.

Equal Opportunity

Gap in recall (TPR) — among true >\$50K earners.

FPR Gap

Gap in false-positive rate among true $\leq$ \$50K earners.

Fairness gaps across audited attributes (percentage points)

Attribute	Statistical Parity Gap	Equal Opportunity Gap	FPR Gap
Sex	16.7	10.3	6.2
Race	15.6	23.8	7.7
Workclass	41.5	28.5	23.6
Age group	30.7	43.0	12.8
Marital status	37.9	46.8	18.6
Education group	68.6	74.6	34.9

The ADS does not satisfy statistical parity, equal opportunity, or FPR parity.

Largest disparities: education, marital status, age group, and workclass

Dropping race ≠ a race-neutral model



INTERSECTIONAL FNR

Intersectional FNR: Race × Education

Intersectional group	FNR
White Less-than-HS	83.7%
Black Less-than-HS	80.0%
Asian-Pac-Islander HS-grad	75.0%
White HS-grad	72.7%
White Graduate	9.0%

Disparities compound when attributes combine

Single-attribute analysis can hide severe subgroup errors. In the race × education audit, lower-education intersectional groups have FNRs above **70%**, while White Graduate individuals have only **9.0%** FNR.



PROXY ANALYSIS — CRAMÉR'S V vs. RACE

Removing race ≠ race-neutral model

Feature	Cramér's V
native.country	0.408
sex	0.117
relationship	0.097
marital.status	0.083
occupation	0.075
education	0.072



Fairness through unawareness fails

Removing the race feature does not make the model race-neutral. Race-related information remains partially encoded through other variables, especially **native.country**, and race-based fairness gaps remain after race is dropped.



Imputation Strategy

Mode imputation hides patterned missingness

- The original implementation replaces hidden “?” values in **workclass** and **occupation** with the most frequent categories.
- But missingness is not random: rows with at least one original “?” have a much lower **positive base** rate than complete rows: **13.3% vs 25.1%**

We compare three imputation strategies:

Imputation Strategy	<i>n</i>	Accuracy	Recall/TPR	FNR
Mode imputation	32,395	85.0%	58.9%	41.1%
Missing as category	32,395	85.3%	59.3%	40.7%
Row deletion	30,042	84.8%	61.2%	38.8%

Treating missing values as an explicit Unknown category is more transparent and slightly improves performance without removing records.



Age-filter audit

166 older individuals removed for no benefit

The original notebook applies an IQR-based age filter that removes older individuals.

The excluded rows are a real older subpopulation:

Group	<i>n</i>	Mean Age	Age Range	> 50K Rate
Kept by age filter	32,395	38.4	17–77	24.1%
Excluded by age filter	166	83.1	78–90	18.1%

Removing the filter slightly improves performance:

Setting	<i>n</i>	Accuracy	Recall/TPR	FNR	ROC-AUC
Author age filter	32,395	85.0%	58.9%	41.1%	0.904
No age filter	32,561	85.6%	60.0%	40.0%	0.910

The age filter narrows the represented population and does not improve model quality. Older individuals should be kept and evaluated separately.

Because the baseline ADS uses **Logistic Regression**, we can inspect the learned coefficients and understand which features push predictions toward **>50K** or **≤50K**.

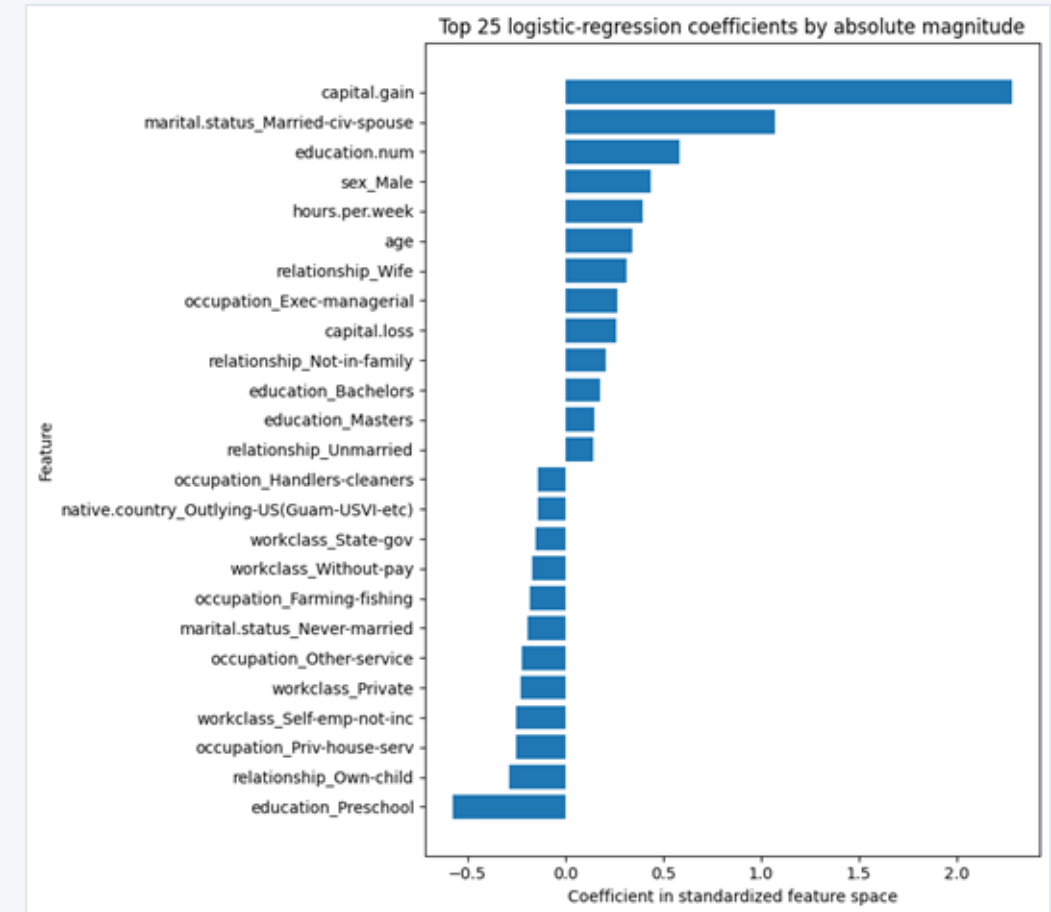
The model tends to assign higher-income predictions to individuals with signals such as: **capital gains, higher education, longer work hours, older age, male sex, and married-civilian-spouse status**.

It tends to lower the probability of **>50K** for individuals associated with: **lower education, younger dependent status, never-married status, some service occupations, and some workclass categories**.

Transparency comes with a performance trade-off.

Logistic Regression is easier to interpret than more complex models, but it may achieve lower predictive performance than less transparent models.

However, because income prediction is socially consequential, transparency should be treated as a design goal, not a secondary concern.

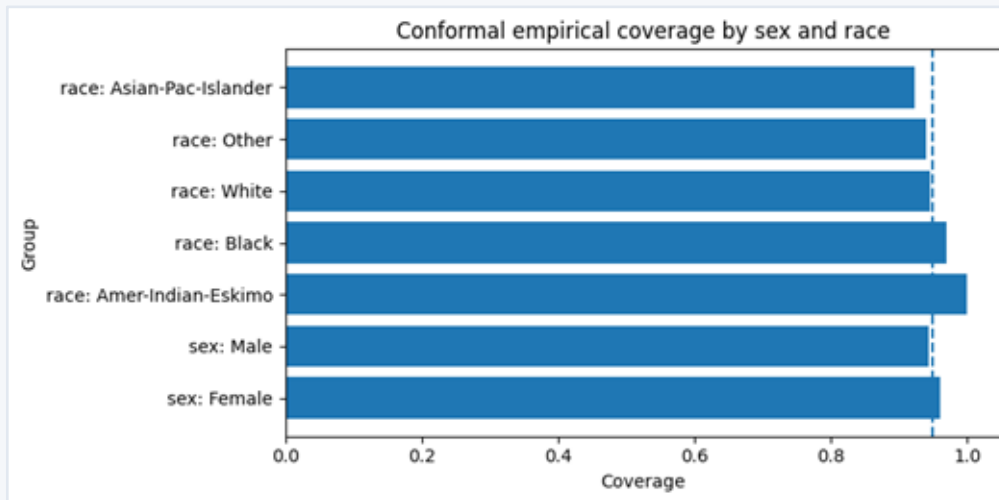




Conformal prediction

94.8% overall coverage, but unequal by group

Average prediction set size = 1.26; 74.4% singleton sets



Coverage falls below the 95% target for some groups, especially **Asian-Pac-Islander individuals: 92.4%, Bachelors: 93.6%, and Graduate individuals: 90.1%.**



Privacy risk

36.8% of records uniquely identifiable

The dataset contains sensitive demographic and employment-related features, including **age, education, marital status, occupation, race, sex, native country, and income.**

The quasi-identifier audit uses: **age + education + marital status + occupation + race + sex + native country**

Privacy metric	Value
Median equivalence class size	3
Unique records	36.8%
Equivalence class size below 5	61.9%
Equivalence class size below 10	77.2%

These results do not prove re-identification, but they show that many records are highly specific.

A responsible deployment should generalize rare values, avoid releasing row-level records, and consider differential privacy for public model outputs or statistics.

SUMMARY & RECOMMENDATIONS

Not ready for high-stakes deployment.

Reasonable aggregate accuracy hides uneven subgroup performance, unjustified preprocessing choices, unequal uncertainty coverage, and limited privacy protection.

KEY FINDINGS

- **High accuracy hides asymmetric errors:** FN = 664 vs
FP = 321
- **Sex- and race-based disparities persist**
- **Education shows the largest fairness gap:** 74.6 pp
- **Intersectional FNRs reach 80%+**
- **Race proxies survive feature removal**
- **Unequal uncertainty coverage and privacy risk remain**

RECOMMENDATIONS

- **Treat “?” as explicit Unknown category**
- **Remove or justify the IQR age filter**
- **Report subgroup metrics, not just accuracy**
- **Retain sensitive attributes for internal fairness auditing**
- **Report subgroup conformal coverage, not only overall coverage**
- **Generalize rare quasi-identifiers; consider differential privacy for public release**